

Track 2: SBIR-Style Translational / Technology Proposal

Project Title

Agent-Based Modeling for Biofilm-Resistant Medical Device Surface Design

Project Summary

Medical device-associated infections remain a major clinical challenge because bacteria can attach to the surfaces of catheters, implants, and prosthetics and form biofilms that are much harder to eliminate than free-floating cells (Donlan, 2001; Hall-Stoodley et al., 2004). This project proposes a computational design tool that uses agent-based modeling to predict how catheter surface properties, including roughness, hydrophobicity, antimicrobial coating strength, and fluid shear conditions, influence bacterial colonization and biofilm formation. Using MESA in Python (ter Hoeven et al., 2025), we modeled individual bacterial agents undergoing planktonic motion, probabilistic surface attachment, EPS secretion, detachment under shear, nutrient-driven growth, and surface spreading on an idealized catheter surface. The model outputs include steady-state attachment rate, EPS surface coverage, biomass, and relative material ranking. Proof-of-concept simulations showed that shear flow and antimicrobial coating strength reduced predicted attachment, while roughness increased biofilm risk. The intended impact is to equip medical-device engineers with a low-cost early-stage tool for comparing surface design choices before physical prototyping, helping guide the development of safer medical devices with lower infection risk.

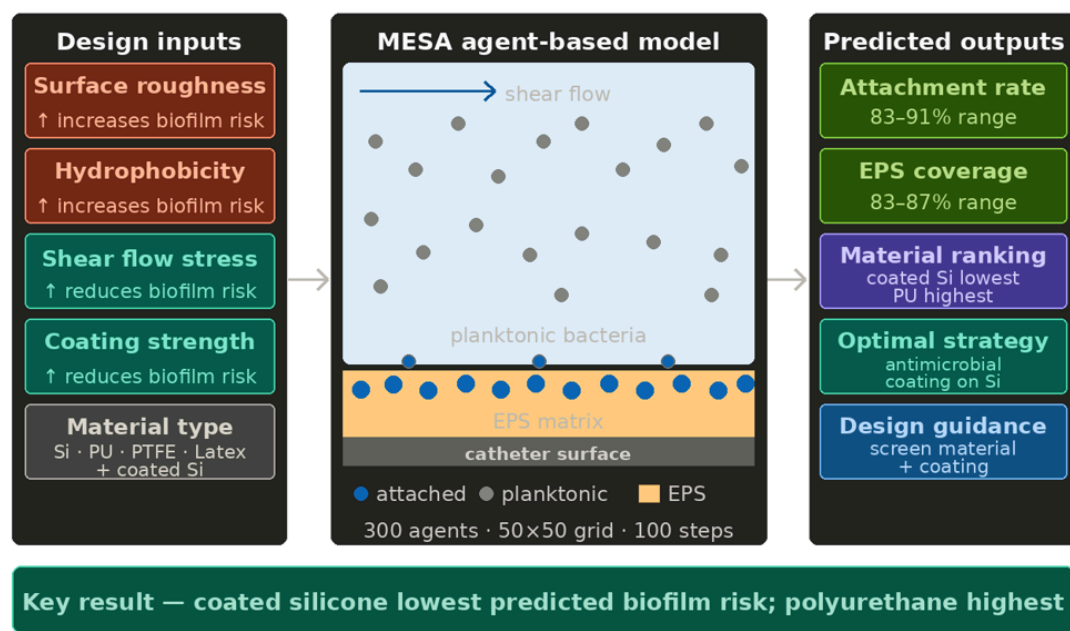


Fig. 1. Graphical abstract of the proposed design-screening workflow. Design

inputs such as surface roughness, hydrophobicity, shear flow, coating strength, and material type are passed into a MESA agent-based model, which predicts attachment rate, EPS coverage, material ranking, and an optimal design strategy.

Project Description

Intellectual Merits

Unmet need. Infections associated with medical devices cause significant patient harm and require costly device removal. The underlying biology of biofilm formation is well studied, and existing computational tools such as iDynoMiCS and NUFEB demonstrate that individual-based biofilm modeling is already an established research approach (Lardon et al., 2011; Li et al., 2019). However, these platforms are designed primarily as microbiology research tools rather than fast engineering design-screening tools. No existing open tool treats surface roughness, hydrophobicity, antimicrobial coating strength, and fluid shear as co-varied design inputs with outputs that directly inform fabrication and design decisions.

Proposed innovation. We built a parametric agent-based model using MESA in Python, treating each bacterium as an independent agent with rules governing surface attachment, planktonic random walk, EPS secretion, detachment under shear, and cell division (Fig. 1). An important design choice was the EPS feedback loop: as attached bacteria secrete extracellular polymer, the local EPS concentration increases the probability that nearby planktonic cells attach (cell-to-cell adhesion) and decreases the probability that established cells detach (biofilm matrix protection). In contrast to uniform surface coverage methods, this creates spatially heterogeneous biofilm patches that more closely resemble experimentally observed biofilm matrix behavior (Flemming & Wingender, 2010).

Computational approach. The model represents a 50 x 50 catheter surface grid, starting with 300 initial planktonic bacteria. Each timestep, free bacteria perform a shear-biased random walk and attempt surface attachment using a probability function incorporating roughness, hydrophobicity, coating effect, local EPS, and shear flow. Next, attached bacteria deposit EPS, compete for energy via nutrient uptake, and divide when energy crosses a threshold. We ran three analyses: (1) 2D parameter sweep over roughness x coating strength and roughness x shear flow (Fig. 2, Fig. 3), evaluating steady-state attachment rate across 64 simulations per sweep; (2) one-at-a-time sensitivity analysis using a central-difference approximation to rank the influence of each design parameter individually (Fig. 4); and (3) material comparison with literature-motivated parameter values for latex, polyurethane, PTFE, uncoated silicone, and antimicrobial-coated silicone, averaged over three stochastic replicates (Fig. 5).

Model logic and parameter implementation. The computational core of the project is a rules-based agent-based model that treats each bacterium as an individual agent instead of part of a continuous population field. This modeling choice takes advantage of the fact that early biofilm formation is spatially heterogeneous: a small number of bacteria attach

first, secrete extracellular polymeric substances, and then create local regions where future attachment becomes more likely. For this simulation, the catheter surface is represented as a 50 x 50 grid, and 300 initial bacteria move across or near the surface as individual agents. Each bacterium has a position, attachment state, energy level, and local interaction with the surrounding EPS field.

In the current ABM, roughness is treated as a normalized relative parameter rather than a directly measured physical quantity. We manually assigned values such as 0.05 for smooth silicone, 0.20 for typical silicone, and 1.00 for a rough comparison surface so the model could compare how increasing roughness changes bacterial attachment and biofilm persistence.

To compare the attachment rates of different material, properties including roughness, hydrophobicity, and coating effect are then defined for each material as their properties from the following ways:

- Silicone and Polyurethane
 - Silicone was assigned the smoother baseline and polyurethane a moderately higher roughness based on literature showing that polyurethane tube surfaces are rougher than silicone tube surfaces and that silicone behaves as the more hydrophobic material (Lima et al., 2011; Palchesko et al., 2012).. The exact normalized values used in the ABM were then chosen as proof-of-concept design inputs rather than direct physical conversions from a single experiment.
- Polyurethane
 - Polyurethane was assigned a moderately higher roughness and lower hydrophobicity than silicone based on reported tube roughness comparisons and a polyurethane water contact angle of about 91° (Lima et al., 2011; Xu et al., 2017).
- PTFE (Teflon)
 - PTFE was treated as a highly hydrophobic uncoated surface. Its roughness was set to 0.05 as a smooth proof-of-concept baseline, since the current model did not use a single catheter-specific AFM roughness value for PTFE and instead treated roughness as a normalized design variable. Its hydrophobicity was set to 1.0, based on literature reporting a shear flow contact angle of about 118 to 120° for pristine PTFE. Its coating effect was set to 0.00 because this case represents bare PTFE rather than a coated surface (Bera et al., 2018; Kang et al., 2023; Mondal et al., 2020).
- Latex
 - Latex was included as an uncoated comparison material. In the current proof-of-concept model, its roughness was set to 0.05 and its hydrophobicity was set to 0.80 as reasonable normalized assumptions rather than direct one-to-one conversions from a single measured literature value. This was done because we did not identify one clean

source in the model setup that provided a matched catheter-specific latex roughness and contact-angle pair comparable to the silicone and polyurethane inputs. Its coating effect was set to 0.00 because the surface was treated as uncoated. This means the latex values should be interpreted as approximate design-screening inputs rather than experimentally calibrated material constants (Pandey et al., 2019).

To represent the fact that fluid motion can transport cells along the surface and reduce residence time near any one location, during each time step, planktonic bacteria perform a random walk that is biased by shear flow. In this simulation, shear flow represents the effect of fluid motion near the catheter surface rather than a fully resolved cleaning or flushing process. After moving, the free bacteria then evaluate whether they attach to the surface using a probability function that combines surface roughness, hydrophobicity, coating strength, shear flow, and local EPS concentration. Higher roughness increases attachment because microscale surface features can protect bacteria from removal, especially from shear flow, and therefore provide more favorable attachment sites. Hydrophobicity is treated as a simplified surface chemistry variable that increases attachment tendency. Antimicrobial coating strength represents either bactericidal activity or anti-adhesive surface treatment which reduces attachment probability. Shear flow reduces initial attachment because higher flow makes it harder for cells to remain near the surface long enough to become sessile, consistent with prior work showing that hydrodynamics influence biofilm structure and attachment behavior (Stoodley et al., 1999).

Once a bacterium attaches, it converts from a planktonic cell to a sessile, surface-associated cell. Attached cells deposit EPS into their local grid location, which creates a positive feedback loop. A higher EPS value increases the probability that nearby planktonic bacteria will attach and also decreases the probability that bacteria that are already attached will detach. This property captures one of the most important features of biofilms because the extracellular matrix helps protect cells and promote collective colonization (Flemming & Wingender, 2010). This model also includes energy accumulation and cell division. The attached bacteria gain energy based on nutrient availability and bacterial growth rate parameters. When energy passes a certain threshold, the cells can divide and create another bacterial agent nearby. This means that biofilm expansion is a function of matrix production, local retention, nutrient-driven growth, and spatial spreading, not just initial attachment. In addition, detachment is modeled probabilistically with shear flow increasing detachment and local EPS reducing it. This property means that a high-shear condition can remove weakly attached cells early in contrast to a mature EPS-rich region, where it becomes harder for shear flow to engage in removal.

The model outputs were chosen to be interpretable for engineering design instead of only biological modeling. The steady-state attachment rate measures the fraction of bacteria that are sessile at the end of the simulation. EPS surface coverage provides an estimate of how much of the catheter-like surface has developed matrix material, which is relevant

because EPS-rich areas are harder to treat and indicate more mature biofilm. For colonization severity, total biomass and attached cell count provide additional measurable data. The parameter sweep outputs are then used to understand and test design trade-offs. The roughness-coating heatmap tests whether antimicrobial coating can compensate for rougher surfaces. The shear-roughness contour tests whether higher flow can reduce biofilm formation even when the surface is less ideal. The sensitivity analysis is asking which single parameter has the largest effect from the baseline condition. Finally, the material comparison converts the abstract parameter model into a practical engineering scenario by assigning normalized values to candidate catheter materials. Altogether, these analyses make the model more than a visualization by turning it into a prototype decision support workflow for comparing surface design choices before physical testing.

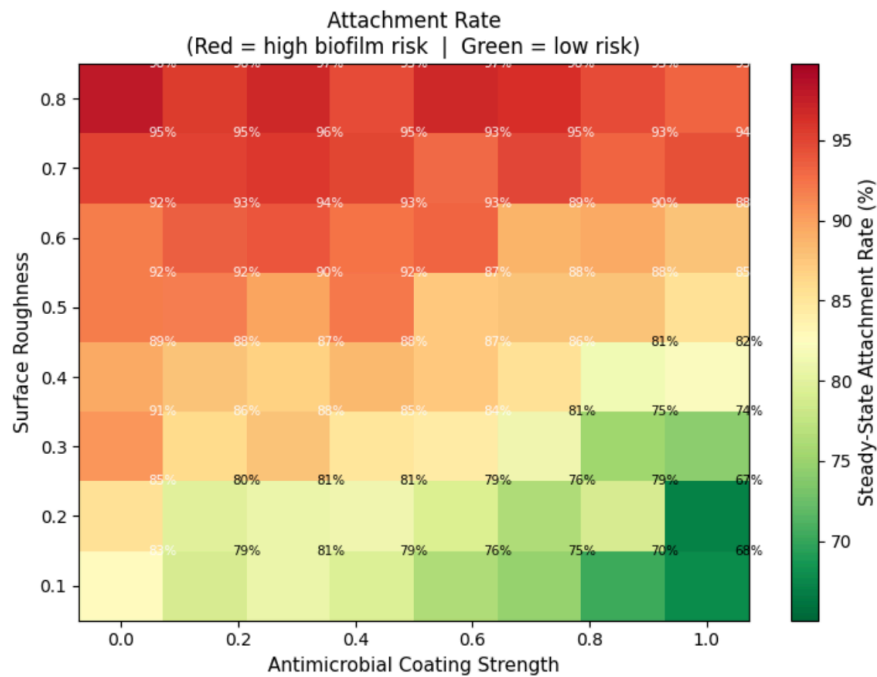


Fig. 2. Surface roughness vs. antimicrobial coating strength heatmap. Steady-state attachment rate as a function of roughness and coating strength. Red indicates higher predicted biofilm risk, while green indicates lower predicted biofilm risk.

Proof-of-concept results. During the baseline silicone-like condition (roughness = 0.10, shear = 0.10, no coating), the model reached a high-attachment state with approximately 86-90% of bacteria in the sessile state and a significant amount of EPS surface coverage, consistent with expectations for a low-flow catheter environment. The sensitivity analysis identified shear flow to be the strongest single protective factor (most negative sensitivity, Fig. 4). Not as impactful in the other direction, surface roughness and

hydrophobicity were the strongest risk-increasing parameters.

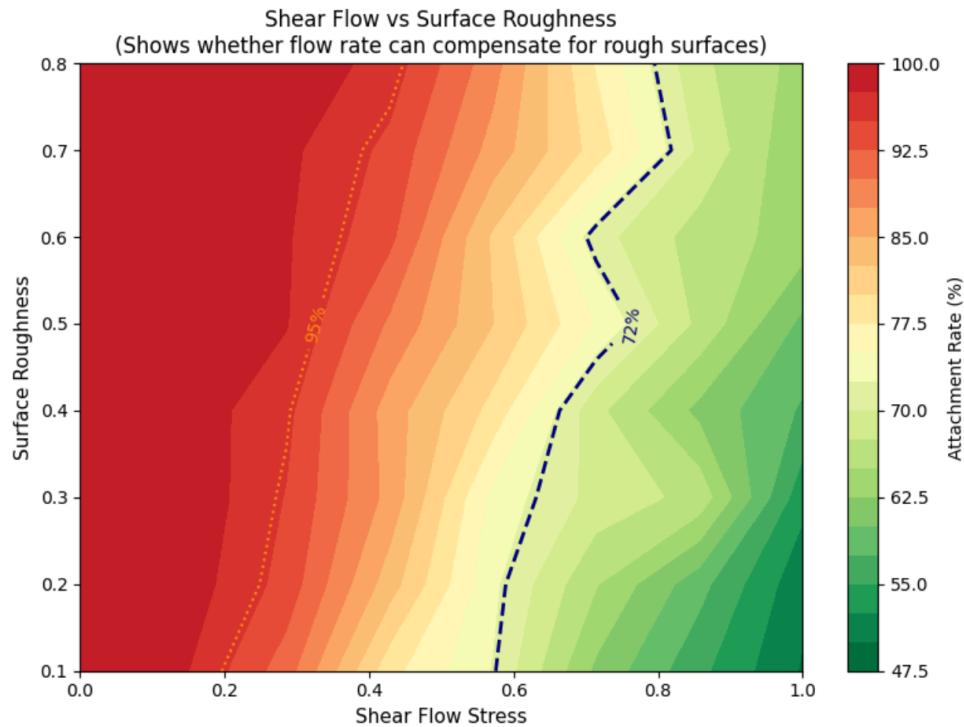


Fig. 3. Shear flow vs. surface roughness contour. Effect of shear flow and surface roughness on predicted steady-state attachment rate. Contour lines mark the lower-risk navy region and upper-risk orange region of the parameter space.

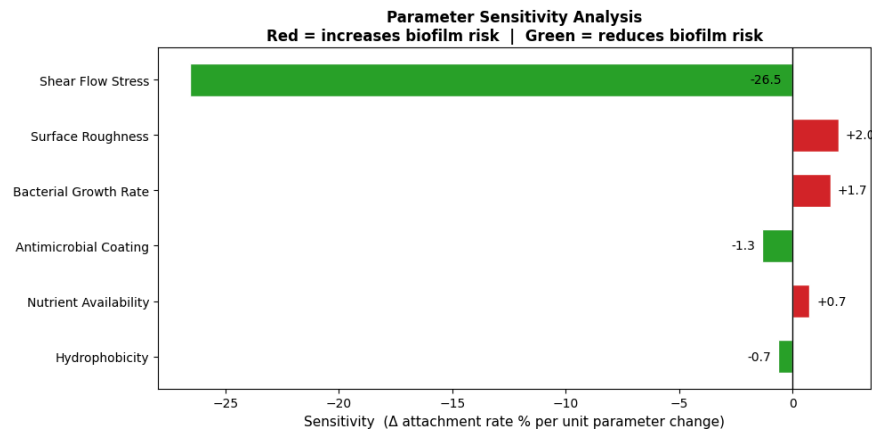


Fig. 4. One-at-a-time parameter sensitivity analysis. Red bars indicate parameters that increase predicted biofilm risk, while green bars indicate protective factors. Shear flow produced the strongest protective effect in the baseline parameter regime.

The material comparison analysis provided a separate proof-of-concept test of whether the ABM could turn abstract design parameters into a practical biomaterial ranking. Five

candidate surfaces were compared: uncoated silicone, polyurethane, PTFE/Teflon, latex, and silicone with an antimicrobial coating. Each material was assigned normalized roughness, hydrophobicity, coating strength, nutrient availability, shear flow, and growth-rate values. Next, to account for random bacterial motion and probabilistic attachment, each condition was averaged across three stochastic simulation runs. The model predicted the silicone with antimicrobial coating to have the lowest steady-state attachment rate, at approximately 83.0%, compared with 87.3% for uncoated silicone. The result supports the direction that increasing antimicrobial or anti-adhesive coating strength lowers the predicted bacterial colonization. The EPS matrix coverage plot depicted that all tested materials developed substantial extracellular polymer coverage, which indicates that once biofilm formation begins, the model tends toward a mature biofilm state under the selected low-flow, catheter-like environment.

The uncoated material ranking should be interpreted more cautiously. The simulation ranked polyurethane as the highest-risk surface, followed by PTFE, uncoated silicone, latex, and antimicrobial-coated silicone. This does not fully match the reference trend used for comparison, where catheter-biomaterial susceptibility was reported to vary across latex, silicone, polyurethane, and other tested materials (Pandey et al., 2019). This mismatch helps identify the need for an important calibration in the current model. The material comparison is currently controlled by simplified scalar inputs, especially for roughness, hydrophobicity, and coating effect. Real bacterial adhesion depends on additional material-specific factors such as surface energy, protein conditioning films, species-specific adhesion molecules, and catheter manufacturing quality. The main feasible result for this material comparison is that the model can compare candidate material conditions and correctly respond to protective design interventions including antimicrobial coating. The next step would be to calibrate the material parameters against experimental catheter biofilm datasets before using the ranking for design decisions.

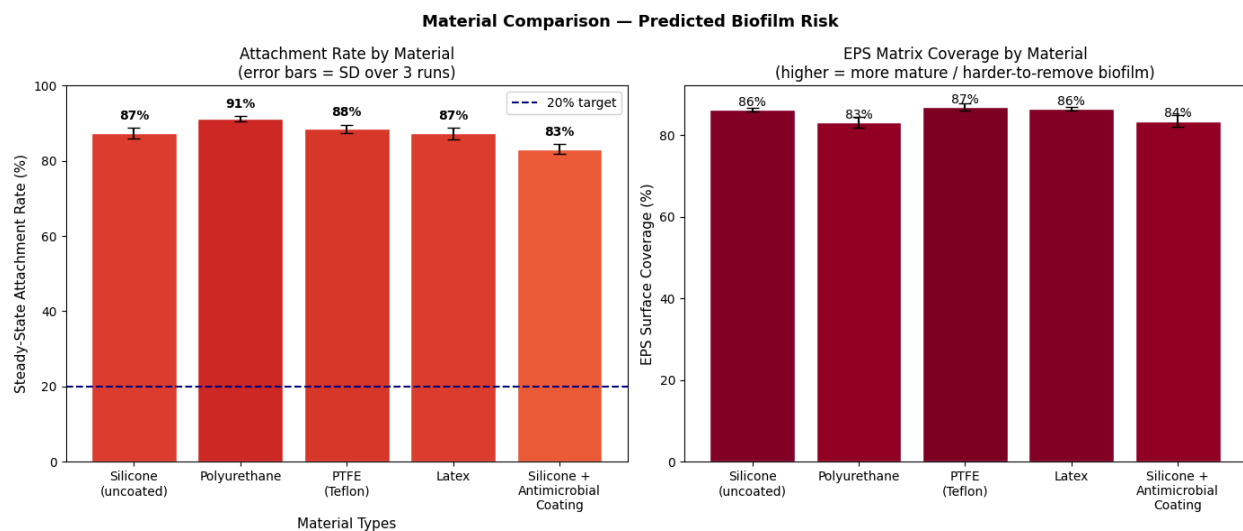


Fig. 5. Material comparison of predicted biofilm risk. Steady-state attachment rate and EPS matrix coverage for uncoated silicone, polyurethane, PTFE/Teflon, latex, and

antimicrobial-coated silicone. The antimicrobial-coated silicone case produced the lowest predicted attachment rate, while the uncoated material ranking requires further calibration against experimental data.

Limitations. The model we built is a proof-of-concept and has a few important limitations. The 2D surface representation chosen for simplicity ignores three-dimensional biofilm tower structure. Shear flow is treated as a single scalar instead of a more accurate spatially resolved fluid velocity field. It would take a coupled FEM-ABM (e.g., FEniCSx + MESA) to model realistic catheter flow geometries. Attachment and detachment probabilities are based on simplified relationships motivated by prior biofilm literature, but they are not calibrated to a specific bacterial species or experimental dataset (Flemming & Wingender, 2010; Stoodley et al., 1999). Nutrient availability is a fixed environmental input instead of a diffusing field. Another limitation is that several parameters are normalized design variables rather than directly measured physical quantities. Overall, this means the model is strongest for comparing trends across design conditions, not for predicting absolute clinical infection rates.

Company/Team

This project is being developed by two UC Berkeley mechanical engineering students with complementary biomedical and computational modeling experience. Nicole is a senior in Mechanical Engineering with experience in generating and analyzing wearable health-technology data for clinical research, which supports the project's connection to medical-device design and patient-facing biomedical applications. Kevin is a second-year Mechanical Engineering student with experience in flight dynamics modeling and computational systems, which supports the implementation of rule-based simulation, parameter sweeps, and model interpretation. The scope is realistic because the goal is not clinical validation, but rather a course-scale proof-of-concept screening workflow that could later be calibrated with experimental catheter biofilm data.

Broader Impacts

This project could benefit patients who rely on catheters and other indwelling or implanted medical devices, as well as clinicians, hospitals, and medical-device designers working to reduce device-associated infections. Biofilm-resistant surface design is important because biofilms can make bacterial contamination more persistent, harder to treat, and more likely to require device replacement or additional clinical intervention (Donlan, 2001; Hall-Stoodley et al., 2004). By connecting computational modeling with biomaterial design, this project supports a more systematic way to evaluate surface roughness, coating performance, and flow conditions before committing to physical prototypes.

The broader scientific value of the project is that it translates biofilm mechanisms into an interpretable engineering model. Rather than only reporting whether a simulated biofilm forms, the ABM identifies which design variables most strongly influence attachment and

EPS coverage. The current results indicate that smoother catheter materials, stronger flow conditions, and effective surface coatings are the most promising design strategies for reducing biofilm formation and lowering infection risk. At the same time, the model should not be interpreted as a clinical decision tool. The material ranking did not fully match the expected literature trend, so this result should be interpreted as a demonstration of the model's screening workflow rather than a final validated ranking of catheter materials. Future versions should incorporate experimental calibration, species-specific parameters, and safety evaluation of any coating strategy before being used in real medical-device development.

Commercialization Potential

The target users for this work are medical-device engineers, biomaterials researchers, and companies developing catheters or other surfaces vulnerable to biofilm formation. The unmet need is for a fast, lower-cost way to screen material and surface-design options before moving into expensive bench testing, coating trials, or prototype fabrication. A realistic first use case would be a pre-prototyping decision-support workflow in which an engineer enters candidate surface parameters including relative roughness, coating strength, hydrophobicity, and expected flow environment, and the tool returns predicted attachment rate, EPS surface coverage, and sensitivity rankings.

The value proposition is that this tool could help reduce the number of weak design candidates that need to be physically tested. For example, a design team could use the ABM to compare whether changing surface roughness, coating strength, or flow conditions is predicted to have the largest effect on biofilm risk. The near-term development pathway would include calibrating attachment and detachment coefficients using published or newly collected catheter biofilm experiments, including catheter-material susceptibility studies such as Pandey et al. (2019). We could also add uncertainty ranges across stochastic replicates and package the model into a simple graphical interface. Over the long term, this ABM could be coupled to a fluid model so that shear stress varies realistically over catheter geometries instead of being treated as a single scalar input.

References

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AI Usage Appendix

AI tools were used to assist with editing, organization, rubric checking, wording refinement, citation placement, and interpretation of report structure. The project idea, code development, computational outputs, figures, and final technical decisions were reviewed and edited by the student team. AI-generated wording was modified for accuracy, consistency with the code outputs, and alignment with the SBIR-style report requirements. The team verified the computational claims by comparing the written interpretation against the notebook outputs, including parameter sweeps, sensitivity analysis, and material comparison results.